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UNDERSTANDING (UN)RELIABILITY OF STEERING VECTORS IN LANGUAGE MODELS

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ABSTRACT

Steering vectors are a lightweight method to control language model behavior by adding a learned bias to the activations at inference time. Although steering demonstrates promising performance, recent work shows that it can be unreliable or even counterproductive in some cases. This paper studies the influence of prompt types and the geometry of activation differences on steering reliability. First, we find that all seven prompt types used in our experiments produce a net positive steering effect, but exhibit high variance across samples, and often give an effect opposite of the desired one. No prompt type clearly outperforms the others, and yet the steering vectors resulting from the different prompt types often differ directionally (as measured by cosine similarity). Second, we show that higher cosine similarity between training set activation differences predicts more effective steering. Finally, we observe that datasets where positive and negative activations are better separated are more steerable. Our results suggest that vector steering is unreliable when the target behavior is not represented by a coherent direction.

1 INTRODUCTION

Activation steering (Turner et al., 2023; Zou et al., 2023) is a promising paradigm for controlling language model outputs using inference-time interventions on model activations. Most works on activation steering have so far focused on steering vectors, which leverage the observation that many human-interpretable behaviors and concepts like truthfulness (Marks & Tegmark, 2024a), refusal (Arditi et al., 2024), and sentiment (Tigges et al., 2023; Konen et al., 2024) are represented as linear directions in models' activations – such that moving in that direction results in greater expression of the given behavior. Steering vector methods control LLM behavior simply by adding a learned bias to the residual stream activations during inference. Although steering vectors were shown to perform well for certain behaviors (Rimsky et al., 2024), recent work demonstrates that steering effects vary significantly across the behaviors, and are often unreliable or even counterproductive (Tan et al., 2024; Brumley et al., 2024; Pres et al., 2024). In this paper, we study Contrastive Activation Addition (CAA) (Rimsky et al., 2024), a representative steering method where the steering vector is computed as the mean difference between activations of datapoints with and without the desired behavior. We evaluate CAA on 36 binary-choice datasets about language model assistant behavior and personality by Rogers et al. (2023), for which previous work finds that CAA performs well for some datasets but not others (Tan et al., 2024).

This paper makes the following contributions:

1. We study how prompt types change the resulting steering vectors and find that the learned vectors differ in direction and steering performance, but all have a positive effect on average.
2. We show that directional agreement, as measured by cosine similarity between activation differences of the training data, is predictive of steering success for the resulting steering vector.
3. We find that separability of positive and negative activations across the steering vector direction is both a conceptual explanation, and empirical predictor for steering success.

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2 METHODS AND EXPERIMENTAL SETUP

Datasets and model. To allow for comparison to prior work, our experiments follow the dataset and model selection of Rimsky et al. (2024) and Tan et al. (2024) by using the 36 multiple-choice datasets by Rogers et al. (2023) and the Llama2-7B chat model (Touvron et al., 2023). Each dataset assesses a different behavior related to language model assistant personality, potentially dangerous behaviors and attitudes towards politics, ethics and more. The 1000 samples per dataset are triples $(x, y^+, y^-) \in \mathcal{D}$, consisting of a prompt x, and answers that match or don't match the steered behavior y^+ and y^-. For instance, if the behavior of interest is honesty, the prompt could be `Is the sky often blue?`, the answer matching the behavior would be `Yes`, and the non-matching answer would be `No`.

Steering Method: Contrastive Activation Addition. We use Contrastive Activation Addition (CAA) by Rimsky et al. (2024) as the steering method. To compute the layer- and behavior-specific steering vector $\mathbf{s}^l \in \mathbb{R}^d$ from training dataset $\mathcal{D}_{\text{train}} = \{(x_i, y_i^+, y_i^-)\}_{i=1}^{\mathbb{N}_{\text{train}}}$, we record residual stream activations at layer l (we use layer l=13 following (Tan et al., 2024)). In the *prefilled* prompt type used in Rimsky et al. (2024); Tan et al. (2024), activations are recorded at the position of the answer token (y^+ or y^-) when it is appended to the prompt. The resulting activations are noted $\mathbf{a}^l(x, y^+))$ and $\mathbf{a}^l(x, y^-))$ respectively. The steering vector $\mathbf{s}^l \in \mathbb{R}^d$ is the mean difference between positive and negative activations: $\mathbf{s}^l = 1/|\mathcal{D}_{\text{train}}| \sum_{\mathcal{D}_{\text{train}}} \left[\mathbf{a}^l(x, y^+) - \mathbf{a}^l(x, y^-)\right]$. To steer during inference, we add $\lambda \mathbf{s}^l$ to the residual stream at layer l. Here $\lambda \in \mathbb{R}$ is the steering multiplier; most of our experiments are done with $\lambda = 1$.

Evaluation of Steering Success. We evaluate steering on a held-out test set $\mathcal{D}_{\text{test}} = \{x_i\}_{i=1}^{\mathbb{N}_{\text{test}}}$ of plain prompts. For each prompt x_i, the model generates an answer token logit distribution, once *with* and once *without* steering. We follow Tan et al. (2024) in using the logit-difference propensity metric: $m_{LD}(x_i) = \text{logit}(y^+) - \text{logit}(y^-)$. We measure steering effect size as $\Delta m_{LD}(x_i) = m_{LD}^{\text{steered}}(x_i) - m_{LD}^{\text{not steered}}(x_i)$, to capture the difference steering makes to the existing model answer propensity. To quantify the reliability, we measure the fraction of anti-steerable samples: $P(\Delta m_{LD}(x_i) < 0)$ for which steering negatively impacts the m_{LD} compared to no steering. Throughout the paper, we use "steerability" and "steerable" to include both the steering effect size and its reliability, diverging from the narrower definition in Tan et al. (2024).

Prompt Variations. We evaluate steering vectors trained using seven prompt types that vary in three components: whether the final answer token is already appended ("prefilled"), whether an instruction is prepended, and whether 5-shot demonstration examples are included. A detailed description of all seven setups, along with an example, are provided in Appendix A. In the *non-prefilled* prompt type, the model is given the prompt x without the answer token appended, and the activations are recorded at the last token position of the prompt (so, while the model generates an answer token). Since we want to get different answers (positive and negative) and the prompt is the same, we prepend instructions and/or 5-shot examples encouraging/discouraging the behavior, which gives positive and negative prompts (x^+ and x^-). We also combine the two strategies and use both prefilled answers (y^+ or y^-) and prompts dis/encouraging the behavior to get $\mathbf{a}^l(x^+, y^+)$ and $\mathbf{a}^l(x^-, y^-))$ – in which case activations are recorded at the answer position. Note that we always use the same test prompt format regardless of the prompt type used for the training data.

3 RESULTS

Effect of Prompt Types on Steering Vectors. We train separate steering vectors for each dataset and prompt type using 250 training samples and 500 evaluation samples. Averaged across all datasets, every prompt type achieves a net-positive shift in the model's logits, and no prompt type clearly outperforms the others (Figure 1). All prompt types also perform similarly to one another on the six datasets where steering vectors perform best – and in this case the results seem slightly less noisy. We also observe that both the steering effect size Δm_{LD} and reliability vary significantly within and between datasets. Similarly to Tan et al. (2024), we observe that for approximately one-third of all samples steering changes the logit-difference in the opposite direction, so the probability of the answer showing the desired behavior decreases. The fraction of such *anti-steerable* samples ranges from 3% to 50% for individual datasets.

Surprisingly, while steering performance is similar and correlated across prompt types, the corresponding steering vectors often do not closely align in activation space: vectors trained on the same samples but with different prompts have pairwise cosine similarities ranging from 0.07 to 0.86 (see Appendix C for more details). These prompt type results reinforce the finding by Tan et al. (2024) that steerability is primarily dataset-dependent: steering performance of different prompt types is similar for the same dataset, and changes in similar ways across datasets. Consequently, we continue to investigate what datasets-specific properties influence steering performance and limit our analysis to the "prefilled" prompt type used in Rimsky et al. (2024); Tan et al. (2024).

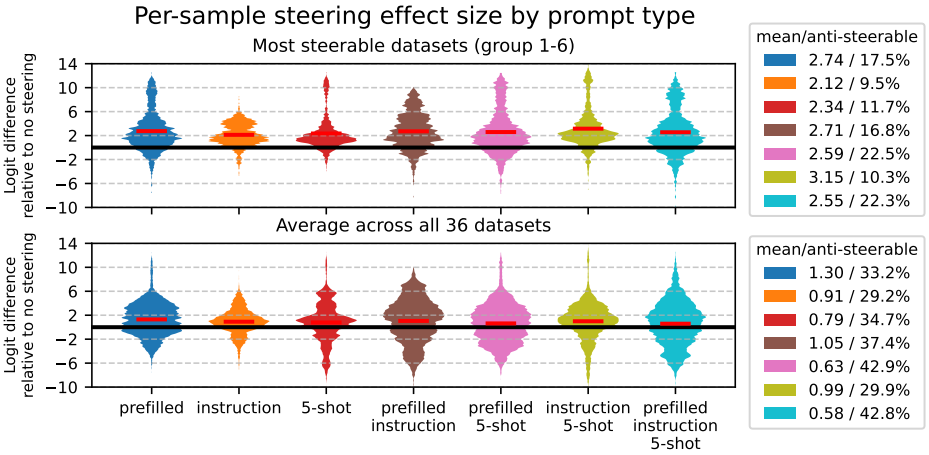

Figure 1: Steering vectors trained with different prompt types all increase the mean logit-difference relative to no steering and perform similarly across datasets. Yet, for all prompt types, steering effect size is unreliable, with a significant fraction of the test samples shifted in the opposite direction ("anti-steerable"). Both steering effect size and faction of such anti-steerable samples vary substantially between datasets, as shown by the six most steerable datasets (top row) outperforming the average shown (bottom row) in both metrics. We used 250 training samples and 500 evaluation samples for each combination of prompt type and dataset.

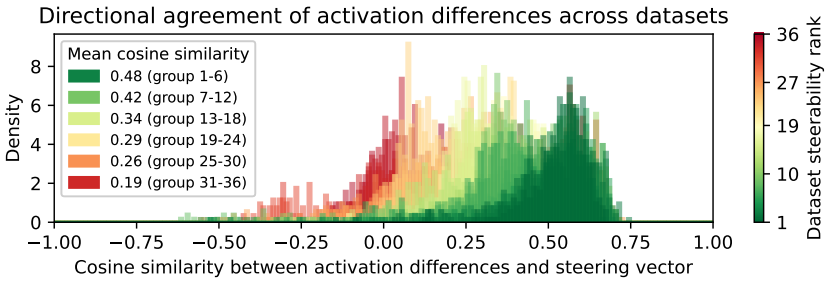

Figure 2: We group the 36 datasets by how effective the resulting steering vector is ("steerability rank"). The most steerable group (ranks 1-6) exhibit high directional agreement between the individual activation differences and the steering vectors, whereas directions in the least steerable group (ranks 31-36) are more dispersed or even orthogonal. Conceptually, high directional agreement suggests a coherent linear representation of the behavior.

Directional Agreement Predicts Steerability. We find that dataset-specific steerability can be explained by directional agreement between the steering vector $\mathbf{s}^l$ and the activation differences $\mathbf{a}^l(x, y^+) - \mathbf{a}^l(x, y^-)$ for the individual data points. If activation differences for a dataset consistently point in a similar direction, this direction approximates the target behavior representation well. Figure 2 shows that datasets with high cosine similarities between activation differences and the steering vector have higher steering vector effectiveness (we order them by their steerability rank from Tan et al. (2024)). We find that higher directional agreement is predictive of both larger steering effect size and fewer anti-steerable samples (see Appendix D for more details). These results provide a concrete explanation for why some behaviors are easier to steer than others. When activation differences for a given behavior align well in activation space, there is a consistent linear direction associated with the behavior represented by the dataset. Conversely, when activation differences are scattered or contradictory, steering vector effectiveness declines.

Difference-of-Means Line Separability Predicts Steerability. By projecting activations onto the *difference-of-means line*, we can assess whether positive and negative activation distributions for a given behavior are naturally separable along the steering direction. We normalize the data such that the mean of positive samples' activations is 1 and the mean of the negative ones is -1. Figure 3 illustrates that for easily steerable behaviors, activations cluster tightly around the means of negative and positive activations, and are fully separable along the difference-of-means-line. For less steerable datasets, however, activation distributions overlap and have high variance along the difference-of-means line. Both directional agreement, as measured by cosine similarity and separability of activations, as measured by the discriminability index d', are correlated with each other and are both predictive of a larger steering effect size and lower fraction of anti-steerable samples.

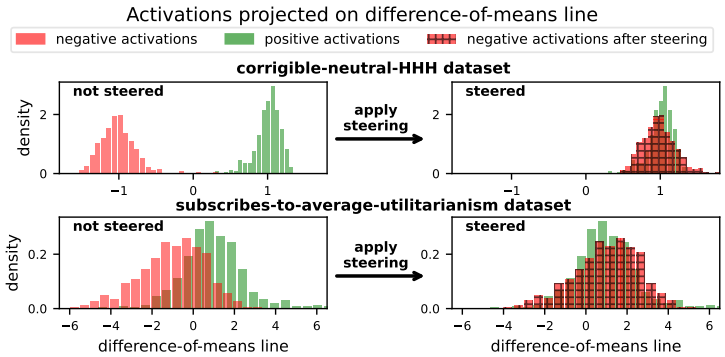

Figure 3: For datasets where the behavior is steerable, activations are clearly separated along the difference-of-activation-means line (top). Less steerable datasets have overlapping positive and negative activations (bottom). CAA steering shifts activations along the difference-of-means line.

4 Discussion

Limitation: breadth of experiments. We evaluate steering performance only using Llama2-7B-Chat on the 36 multiple-choice datasets common in prior work. Future works should investigate different models and non-multiple choice datasets to determine broader generalizability. Further, our study only focused on CAA; while we anticipate that our results will transfer to other steering vector methods like Function Vectors (Todd et al., 2023) and BiPO (Cao et al., 2024), verifying this transfer would be helpful. On the other hand, we are not sure whether our results would generalize to more expressive steering methods such as MiMiC (Singh et al., 2024), ACE (Marshall et al., 2024) or LoREST (Krasheninnikov & Krueger, 2024), all of which involve projection matrices instead of just a shift by a constant vector. Additionally, we believe an investigation into how our prompting strategies affect performance on unrelated general benchmarks like MMLU (Hendrycks et al., 2021) is warranted, as prior work by Stickland et al. (2024) finds that vector steering modestly reduces model performance on downstream tasks.

Limitation: methodology for prompt type comparison. Statistically comparing prompt types is highly sensitive to hyperparameters like training-set size, complicating robust analysis. With few (5–30) randomly sampled training activations, steering vectors for the same prompt type vary so widely that true differences between prompt types are lost in the intra prompt type variance. Conversely, when drawing many (200–500) training activations, the intra prompt type variance disappers (cosine similarity > 0.99), making resampling redundant. While we could run enough subsampling to achieve statistical significance in both cases, we believe this would add little practical insight.

Conclusion. Our work provides a deeper understanding of when and why steering vectors are (un)reliable. First, we find that prompt selection has measurable but limited influence on steering effectiveness, and that no single prompt type consistently outperforms others across datasets. Second, we find that steering vector performance depends on how the target behavior is represented in the activation space. Both directional consistency of activation differences and separability of activations along the difference-of-means line are conceptually intuitive explanations and empirical predictors of steering vector performance. Our results demonstrate that steering vectors are not universally applicable, and that their effectiveness depends on whether the targeted behavior is well-represented as a linear direction in the model's activation space. We hope these insights can inform future methods for more robust and interpretable activation-based interventions.

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